Part I

MASTER'S COMPREHENSIVE EXAMINATION
Theory (90 minutes) December 5, 1992

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. Let the continuous r.v. X have density function $f(x) = |1-x| I_{(0,2)}(x)$. Let A be the event that $1/2 \le X \le 3/2$ and B be the event that $X \ge 1$.

(a) Are A and B independent (show)?

(b) Find $E[|X-1|]$.

2. Let $X_1,...,X_n$ be i.i.d. random variables with density

$$f(x) = \begin{cases} (\lambda/\theta^\lambda)x^{\lambda-1} & \text{if } 0 \le x \le \theta \\ 0 & \text{other.} \end{cases}$$

The unknown parameters λ and θ are such that $\lambda > 0$, $\theta > 0$. Find the form of the most powerful test of

$$H_0: \lambda = 3, \ \theta = 1 \quad \text{versus} \quad H_1: \lambda = 2, \ \theta = 2.$$

3. Suppose X_n has a geometric (p_n) distribution with probability mass function

$$f_n(x) = p_n(1-p_n)^x, \quad x = 0,1,2,\ldots .$$

Furthermore, suppose $np_n \to \lambda$ as $n \to \infty$. Find the limiting distribution of $Y_n = X_n/n$ as $n \to \infty$.

4. Suppose we have the following model

$$Y_t = \beta_1 x_t + \epsilon_t \quad \text{for } t = 1,2,\ldots,n$$

and

$$Y_t = \beta_2 x_t + \epsilon_t \quad \text{for } t = n+1,\ldots,2n,$$

where ϵ_t, $t = 1,2,\ldots,2n$ have independent Normal $(0,\sigma^2)$ distribution with σ^2 unknown, and $x_1,\ldots,x_{2n}$ are known constants. Construct an F–test for testing

$$H_0: \beta_1 = \beta_2 \quad \text{versus} \quad H_1: \beta_1 \ne \beta_2.$$

5. Let X_1 and X_2 be two independent normal observations with a common unknown mean μ and variances $Var(X_1) = 1$ and $Var(X_2) = 2$. Suppose you have the knowledge of $\mu > 0$.

(i) Find an unbiased estimator of μ. (Show the unbiasedness.)

(ii) Find the MLE of μ. (Show how you arrive at the MLE.)

Part II

MASTER'S COMPREHENSIVE EXAMINATION
Applied (90 minutes)

December 5, 1992

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. The following table contains exam scores of a class of 8 male and 7 female students:

Male	60	57	84	69	69	81	66	98
Female	81	69	95	99	66	80	63	

Use the Mann-Whitney rank test to determine whether the performance of female students is significantly better than that of male students at 5% level. What is your null hypothesis?

2. The following table contains data on age versus price of used Dodge vans published in the Classified section of certain newspapers:

Age (yrs)	5	7	6	7	4	7
Price ($1,000's)	7.7	5.7	5.8	5.5	8.8	4.3

(i) Use the least squares method to fit the data with a straight line.

(ii) Suppose the age-price relationship satisfies a simple linear model with normal errors. Is the slope in the model significantly different from -1 at 10% level? Give details of your procedure.

3. To investigate the relative tensile strengths of tape-closed and sutured wounds, two symmetrical incisions on the backs of rats were created. One wound was closed by suture and the other by surgical tape. The animals were sacrificed after 100 days of healing, and the wounds were analyzed mechanically. The results for the healing time of 100 days of the tensile strength (lb/in.^{2}) of taped and sutured wounds for five rats are:

Rat No.	Tape	Suture
1	1627	1571
2	1984	1968
3	926	883
4	1571	609
5	1984	1985

(a) Do the data indicate real differences in tensile strength for tape-closed versus sutured wounds? (Use α = 5%.) Use both a parametric and nonparametric test. Which is more appropriate?

(b) Find a 90% confidence interval for the difference between tape-closed and suture with respect to mean tensile strength.

4. When drinking tea, a British woman claimed to be able to distinguish whether milk or tea was added to the cup first. To test her claim, she was given ten cups of tea, in five of which milk was added first. She was told that there were five cups of each type, so that she should make five predictions of each order. The order of presenting the cups to her was random.

	Guess Poured First	
Poured First	milk	tea
Milk	4	1
Tea	1	4

(a) State the null and alternative hypotheses.

(b) Test the null hypothesis in (a); including a p-value and your statistical conclusions.

5. A pharmacokinetic study was conducted to compare maximum drug uptake for two drug formulations (call them A and B). The study was conducted using six subjects. Each subject was administered both drugs, one month apart. Following each administration drug levels were monitored for 24 hours and the maximum level was determined. Three of the subjects received the drugs in the order A followed by B, the other three subjects received the drugs in the reverse order. The results of this study are shown below.

Subject	Order	Drug A	Drug B
1	2	15	12
2	1	10	10
3	1	8	9
4	2	17	13
5	1	10	12
6	2	17	14

where Order = 1 means drug A followed by drug B
Order = 2 means drug B followed by drug A

Construct the appropriate analysis of variance to determine if the mean response for the two drugs is equal. Be sure to state all hypotheses tested and indicate the conclusions from your analysis.

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Rutgers University
Master's Comprehensive Examination

PART I Theory (90 minutes) March 7, 1992

Instructions: Answer four out of five of the following questions. Indicate which questions you are answering. Open book and open notes.

1) $X_1, X_2, \ldots, X_n$ are i.i.d random variates with density function

$$f(x;\theta) = \theta x^{\theta-1} \text{ if } 0 \le x \le 1 \qquad \theta \ge 1;$$
$$= 0 \text{ otherwise.}$$

a) Find a sufficient statistic for θ.
b) Find the M.L.E. of θ.

2) The random variable X has density function

$$f(x;\theta) = \theta x + 0.5 \quad \text{if } -1 \le x \le 1;$$
$$= 0, \quad \text{otherwise.}$$

a) What are the possible values of θ ?
b) Find the variance of X.
c) Find $P(X^2 > 0.25 \mid X \le 0)$.

3) Components of a certain type are shipped to a supplier in baches of 12. Suppose that 60 percent of all batches contain no defectives; 30 percent contain exactly 1 defective; and 10 percent contain exactly 2 defectives. A sample of three (different) components are selected from the batch at random. Given that the sample contains exactly one defective components, is it more likely that the batch contains exctly one defective as compared to the likelihood of the batch containing exactly two defectives. (Compute and show the probabilities and compare).

4) Given a finite outcome sample space, let A, B, and C be three events. Prove that $P(A \cap B \cap C) \le P(A)\,P(B|A)$.

5) X is a random variable with cumulative distribution function F(x) given below. Find the expected value and variance of X.
$F(x) = 0$, for $x < 1$; $F(x) = 0.2$, for $1 \le x < 3$; $F(x) = 0.6$, for $3 \le x < 7$; $F(x) = 0.9$, for $7 \le x < 11$; $F(x) = 1$, for $11 \le x$.

Rutgers University

Master's Comprehensive Examination

PART II Applied (90 minutes) March 7, 1992

Instructions: Answer four out of five of the following questions. Indicate which 4 you are answering Open book and open notes.

1) Assume that 2 factors A and B have been included in an experiment, each factor at 2 levels. Construct a two-way table (and a graph) using the numbers (24, 37, 57, 70) to illustrate each of the following hypothetical situations. (Note: you may use a number more than once in the same table. Also, no calculations need be performed).

a) No A effect, significant B effect, zero interaction.

b) Significant A effect, significant B effect, zero interaction.

c) No A or B effects, significant interaction.

d) Using the arrangement of the data that you selected in part a) find the residuals assuming the model: $Y_{ij} = \mu + A_i + B_j + \varepsilon_{ij}$, for $i = 1,2$ and $j = 1,2$.

2) A school has 600 students who bought their own computers. You seek to estimate the average cost of these computers. A pilot study estimates that the standard deviation of cost is \$650. How big a simple random sample without replacement should you take if you seek to estimate the mean cost within \$100 with 95% confidence.

3) In a large company 35 percent of the employees are women. Of the employees who are in managerial positions, 25 percent are women. Estimate how much more likely are men to be managers than women to be managers in this company. Explain your estimate and how you computed it. (Hint: assume there are N employees and K managers).

4) In a community, 50 percent of the homes are single family homes, 35 percent are two-family homes, and 15 percents are three-family homes. A review of all homes in the community revealed that 120 homes had problems with radon. Of these, 40 homes were single family homes, 50 were two family homes, and 30 were three-family homes. Is there statistically significant evidence that different types of homes have proportionately more radon than other types of homes. State and carry out test and summarize your results.

5) 200 students take a test that has two questions (Question A and Question B) on it. 60 students get both questions right. 40 students get both questions wrong. 70 students get question A right, but question B wrong. 30 students get question B right, but A wrong. Carry out a test of whether question A is easier than question B.